MANGALORE REFINERY - FIELD INSPECTION REPORT
Report No: INS-2026-0417
Inspection Date: 14 September 2026
Inspector: R. Shetty (Mechanical Integrity)
Unit: Crude Distillation Unit (CDU-2)

Equipment: Heat Exchanger E-2104 (Shell & Tube)
Tag: E-2104A  Service: Crude Preheat

Finding 1: External corrosion observed on shell near the south saddle support, approx. 150 mm x 80 mm. Measured wall thickness: 7.2 mm (nominal 10.0 mm).

Finding 2: Minor weep at channel head gasket (bolt 7). No active leak at time of inspection.

Observation: Insulation damaged over 1.5 m section on inlet nozzle N1; moisture ingress likely.

Equipment: Centrifugal Pump P-2210B
Finding 3: Vibration 7.8 mm/s at drive-end bearing (alarm limit 7.1 mm/s). Bearing temperature 82 C.

Recommendation 1: Carry out UT thickness mapping of E-2104A shell within 30 days; evaluate fitness for service per API 579 if thickness is below 6.5 mm.

Recommendation 2: Replace channel head gasket during next planned shutdown; re-torque bolts meanwhile.

Recommendation 3: Repair insulation on nozzle N1 and inspect for corrosion under insulation (CUI).

Recommendation 4: Schedule P-2210B bearing replacement and laser alignment check within 7 days.

Overall Risk Rating: MEDIUM
Next Inspection Due: March 2027